Experiment no. 02 - Logistic Regression

Aim: To implement Logistic Regression using Python and predict whether a student will Pass (1) or Fail (0) based on study hours, attendance, assignments, and previous marks.

Theory: Logistic Regression is a Supervised Machine Learning Classification Algorithm used to predict categorical outcomes. It is mainly used for binary classification, where the output has only two classes such as:

- Pass / Fail
- Yes / No
- True / False
- Spam / Not Spam

Unlike Linear Regression, which predicts continuous values, Logistic Regression predicts the probability that a data point belongs to a particular class. The predicted probability always lies between 0 and 1 .

Logistic Regression uses the Sigmoid (Logistic) Function to convert the output into a probability.

Sigmoid Function : P=1 /1 +e^-z

where

P = Probability of the positive class

e = Euler's constant (≈ 2.71 8)

If the probability is:

≥ 0.5 → Class = 1 (Pass)

< 0.5 → Class = 0 (Fail)

In this experiment, the model predicts whether a student will pass based on:

- Study
- Hours

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- Attendance
- Assignments
- Previous Marks

The dataset is divided into training data (80%) and testing data (20%). The Logistic Regression model is trained using the training data, and its performance is evaluated using Accuracy.

A graph is plotted to visualize the relationship between Study Hours and the Probability of Passing.

Applications:

1. Student Result Prediction
2. Email Spam Detection
3. Disease Diagnosis
4. Credit Card Fraud Detection
5. Customer Churn Prediction
6. Loan Approval Prediction

Advantages:

1. Simple and easy to implement.
2. Fast training and prediction.
3. Works well for binary classification problems.
4. Produces probability values between 0 and 1 .
5. Easy to interpret.

Disadvantages:

1. Suitable mainly for classification problems.
2. Cannot capture complex non-linear relationships without feature engineering.
3. Performance depends on the quality of the dataset.
4. Sensitive to outliers and highly correlated features.

Dataset (CSV File):

Study_Hours,Attendance,Assignments,Previous_Marks,Pass

1,40,31,30,0

2,45,23,35,0

2,55,42,45,0

3,50,24,42,0

3,65,49,50,1

4,60,34,48,0

4,70,35,58,1

5,68,45,51,1

5,75,66,61,1

5,75,60,60,1

6,70,57,70,1

6,80,65,65,1

7,75,76,68,1

7,85,72,72,1

8,82,70,70,1

8,90,98,82,1

9,88,78,71,1

4,55,44,46,0

5,60,54,60,1

6,65,59,71,1

7,72,64,1

Code:

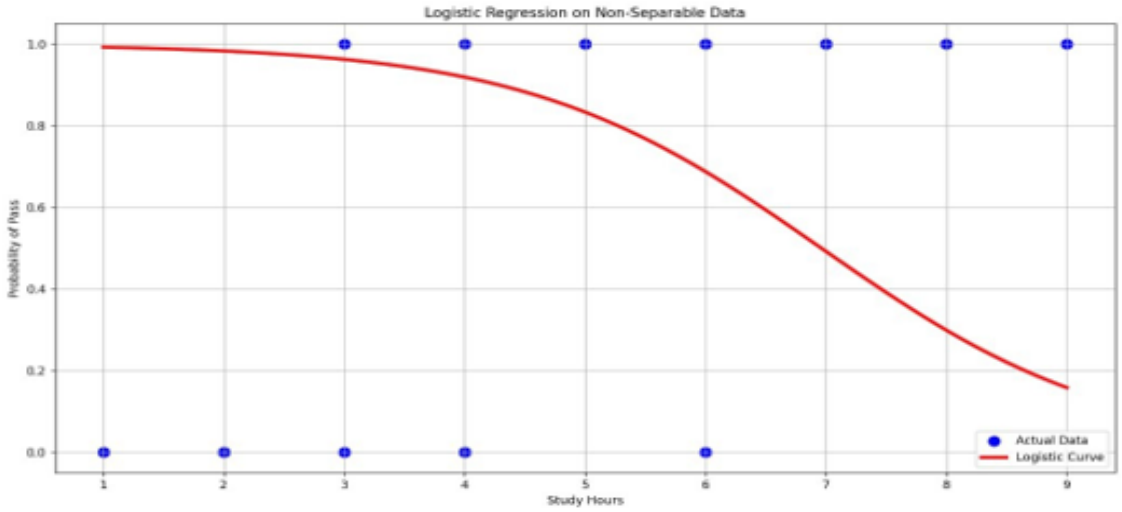
```
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score
# Read CSV File
data = pd.read_csv("Student.csv")
# Input and Output
X = data[["Study_Hours", "Attendance", "Assignments", "Previous_Marks"]]
y = data["Pass"]
# Split Data
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)
# Create Model
model = LogisticRegression()
# Train Model
model.fit(X_train, y_train)
# Prediction
y_pred = model.predict(X_test)
# Accuracy
print("Accuracy =", accuracy_score(y_test, y_pred))
```

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```
# Predict New Student
new_student = pd.DataFrame(
    [[6, 80, 6, 65]],
    columns=["Study_Hours", "Attendance", "Assignments", "Previous_Marks"])
prediction = model.predict(new_student)
probability = model.predict_proba(new_student)
print("Prediction =", prediction)
print("Probability =", probability)
# -------- Graph --------
hours = np.linspace(1, 9, 200)
graph_data = pd.DataFrame({
    "Study_Hours": hours,
    "Attendance": np.repeat(data["Attendance"].mean(), 200),
    "Assignments": np.repeat(data["Assignments"].mean(), 200),
    "Previous_Marks": np.repeat(data["Previous_Marks"].mean(), 200)
})
prob = model.predict_proba(graph_data)[:, 1]
plt.figure(figsize=(8,5))
plt.scatter(data["Study_Hours"], data["Pass"], color="blue", s=80, label="Actual
Data")
plt.plot(hours, prob, color="red", linewidth=3, label="Logistic Curve")
plt.xlabel("Study Hours")
plt.ylabel("Probability of Pass")
```

```
plt.title("Logistic Regression on Non-Separable Data")
plt.grid(True)
plt.legend()
plt.show()
```

Output:



Explanation of the Graph:

Title:

Logistic Regression on Non-Separable Data

This graph shows how Logistic Regression predicts the probability that a student will pass based on study hours.

X-Axis (Horizontal)

Study Hours

It represents the number of hours studied by each student.

Y-Axis (Vertical)

Probability of Pass

The values range from 0 to 1 .

0 = Fail

1 = Pass

Blue Dots

The blue dots represent the actual student data.

Blue dot at 0 → Student Failed.

Blue dot at 1 → Student Passed.

Because this is non-separable data, some students who studied more still failed, while some students who studied less passed.

- Red Curve

The red line is the Logistic Regression (Sigmoid) Curve.

It shows the predicted probability of passing.

If the curve is close to 1 , the probability of passing is high.

If the curve is close to 0, the probability of passing is low.

Conclusion: